

# **Beliefs about Refugee Integration in Germany: A Survey Experiment on Ukrainian and Syrian Refugees**

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# Contents

|          |                                                                          |           |
|----------|--------------------------------------------------------------------------|-----------|
| <b>1</b> | <b>Introduction</b>                                                      | <b>1</b>  |
| <b>2</b> | <b>Literature Review and Conceptual Framework</b>                        | <b>1</b>  |
| 2.1      | Immigration Attitudes and Migrant Characteristics . . . . .              | 1         |
| 2.2      | Beliefs and Expectations about Migrant Outcomes . . . . .                | 2         |
| 2.3      | Differential Perceptions of Ukrainian and Syrian Migrants . . . . .      | 3         |
| 2.4      | Belief Formation Through Contact, Experience, Information, and Memory    | 5         |
| <b>3</b> | <b>Experimental Design</b>                                               | <b>7</b>  |
| 3.1      | Target Population and Proposed Sample . . . . .                          | 7         |
| 3.2      | Factorial Vignette Design . . . . .                                      | 8         |
| 3.3      | Outcome Variables . . . . .                                              | 10        |
| 3.4      | Respondent Questionnaire . . . . .                                       | 11        |
| 3.4.1    | Personal Contact and Experience . . . . .                                | 11        |
| 3.4.2    | Media Exposure and Information . . . . .                                 | 13        |
| 3.4.3    | General Beliefs About Integration . . . . .                              | 13        |
| 3.4.4    | Respondent Controls . . . . .                                            | 15        |
| 3.5      | Data Quality and Processing . . . . .                                    | 15        |
| <b>4</b> | <b>Empirical Strategy</b>                                                | <b>16</b> |
| 4.1      | Experimental Analysis: Identification and Tests of H1–H3 . . . . .       | 16        |
| 4.1.1    | Variables and Baseline Specification . . . . .                           | 16        |
| 4.1.2    | Main Effects: H1a, H1b, H2, H3a, and H3b . . . . .                       | 18        |
| 4.1.3    | Interaction Effects: H3c and Additional Analysis . . . . .               | 20        |
| 4.2      | Associational Analysis: Contact, Experience, Media, and Memory . . . . . | 21        |
| 4.2.1    | Contact and Experience: H4a–H4b . . . . .                                | 21        |
| 4.2.2    | Media Exposure and Tone: H5 . . . . .                                    | 22        |
| 4.2.3    | Memorable Experiences and Information: H6a–H6b . . . . .                 | 23        |
| 4.3      | General Beliefs and Exploratory Analyses . . . . .                       | 24        |
| <b>5</b> | <b>Limitations and Extensions</b>                                        | <b>26</b> |
| 5.1      | Extensions . . . . .                                                     | 26        |
| 5.2      | Limitations and Identification Constraints . . . . .                     | 26        |
| <b>6</b> | <b>Conclusion</b>                                                        | <b>26</b> |
|          | <b>References</b>                                                        | <b>27</b> |
| <b>A</b> | <b>Appendix Survey Instrument</b>                                        | <b>33</b> |
| A.1      | Vignette experiment and profile construction . . . . .                   | 33        |
| A.2      | Outcome questions and response scales . . . . .                          | 33        |

|          |                                                                           |           |
|----------|---------------------------------------------------------------------------|-----------|
| A.3      | Personal contact and experience questions . . . . .                       | 33        |
| A.4      | Media exposure and information questions . . . . .                        | 33        |
| A.5      | General-belief questions . . . . .                                        | 33        |
| A.6      | Respondent characteristics . . . . .                                      | 34        |
| <b>B</b> | <b>Appendix Data Processing</b>                                           | <b>34</b> |
| B.1      | Data-quality procedures . . . . .                                         | 34        |
| B.2      | Missing, not-applicable, and structurally unavailable responses . . . . . | 34        |
| B.3      | Construction of derived variables . . . . .                               | 34        |
| B.4      | Analysis-sample availability . . . . .                                    | 36        |

# 1 Introduction

## 2 Literature Review and Conceptual Framework

### 2.1 Immigration Attitudes and Migrant Characteristics

A large literature examines how host-country populations evaluate migrants and which characteristics shape these evaluations. One central distinction in this literature is based on whether attitudes toward migrants primarily reflect individuals' own economic interests and labor-market considerations or characteristics of the migrants.

Earlier economic studies provide some evidence that labor-market considerations are associated with immigration preferences (e.g., Scheve and Slaughter, 2001; Mayda, 2006). Beyond economic self-interest, immigration attitudes have also been linked to broader fiscal, cultural, and group-related concerns (Burns and Gimpel, 2000; Dustmann and Preston, 2007; Sides and Citrin, 2007). Reviewing this literature, Hainmueller and Hopkins (2014) conclude that evidence for explanations based on individual economic self-interest is generally limited.

Experimental evidence further challenges a simple labor-market competition explanation and rather points to the importance of migrants' own characteristics. Hainmueller and Hiscox (2010) find that both high- and low-skilled natives prefer high-skilled over low-skilled immigrants. If immigration attitudes were primarily driven by competition with similarly skilled workers, preferences would be expected to differ according to respondents' own skill levels. Instead, the common preference for high-skilled immigration suggests that respondents place considerable weight on migrants' human capital and potential economic contribution. This pattern is also found in multidimensional experiments. Higher education, occupational status, employability, and language proficiency are generally associated with more favorable evaluations of migrants (Hainmueller and Hopkins, 2015; Bansak et al., 2016; Czymara and Schmidt-Catran, 2016). More recent evidence from Germany similarly shows greater willingness to accept skilled economic migrants with recognized professional qualifications, good language skills, and a definite job offer (Wolff et al., 2025). Taken together, these studies show that characteristics associated with migrants' expected labor-market integration play an important role in how they are evaluated.

Country of origin, cultural and religious background, and migration status also influence attitudes when other characteristics are held constant. Hainmueller and Hopkins (2015) find differences in evaluations across immigrants' countries of origin alongside effects of education, occupation, and English proficiency. In the European context, Bansak et al. (2016) show that preferences over asylum seekers depend jointly on country of origin, religion, employability, and vulnerability. Similar patterns emerge in Germany. Czymara and Schmidt-Catran (2016) find that acceptance varies with migrants' country of origin and reason for migration as well as their education and German-language proficiency.

These studies show that evaluations depend not only on migrants’ individual characteristics, but also on broader group-related characteristics and migration categories.

Although much of this literature focuses on attitudes and preferences toward migrants rather than beliefs about their expected outcomes, it shows that evaluations vary systematically with migrants’ observable characteristics. In line with this literature, higher human capital and stronger language proficiency may similarly serve as signals when respondents form expectations about migrants’ economic and social integration. This leads to the first hypotheses:

H1a: Holding other migrant characteristics constant, respondents expect migrants with higher human capital to achieve more favorable labor-market integration outcomes than migrants with lower human capital.

H1b: Holding other migrant characteristics constant, respondents expect migrants with higher German-language proficiency to achieve more favorable labor-market and social-integration outcomes than migrants with lower German-language proficiency.

## **2.2 Beliefs and Expectations about Migrant Outcomes**

A growing literature shows that individuals hold systematic beliefs about immigrants and their economic and social characteristics. Much of this research examines perceptions of immigrant populations as a whole, including their size, composition, education, and employment (Sides and Citrin, 2007; Alesina et al., 2023). Respondents tend to overestimate the share of immigrants living in their country, and perceive immigrants as less educated, more unemployed, and more dependent on government transfers than they actually are. However, these perceptions may depend on which types of migrants respondents have in mind. Blinder (2015) shows that public understandings of the term “immigrant” do not necessarily correspond to the full migrant population, with some migrant categories being considerably more salient than others. This suggests that treating migrants as a single category may obscure differences in how distinct types of migrants are perceived.

One particularly relevant distinction concerns migrants’ reasons for migration. Evaluations vary between refugees and migrants arriving for economic reasons (Czymara and Schmidt-Catran, 2016). Previous research on refugee labor-market integration examines several dimensions of migrants’ economic outcomes, including employment, earnings, and occupational outcomes (Brell et al., 2020; Fasani et al., 2024). These studies find that refugees tend to experience less favorable outcomes along these dimensions than economic migrants. Although this evidence concerns evaluations or actual outcomes rather than beliefs about future integration, it provides reason to expect that host-country populations may also distinguish between otherwise comparable refugees and economic migrants when forming such expectations. This leads to the following hypothesis:

H2: Holding all other migrant characteristics constant, respondents expect refugees to achieve less favorable labor-market integration outcomes than economic migrants.

A related literature examines migrants' employability and labor-market experiences and shows that employment outcomes depend not only on migrants' individual characteristics, but also on organizational and institutional factors in the host country (Shirmohammadi et al., 2019; Farashah et al., 2023). Research on immigrant recruitment further demonstrates that employers' perceptions can shape migrants' employment opportunities and that migrant workers may face discrimination and disadvantages in hiring and occupational outcomes (Almeida et al., 2015; Dietz et al., 2015). However, this literature primarily examines migrants' actual labor-market experiences or employers' recruitment decisions rather than expectations held by the broader host-country population.

A smaller literature comes closer to examining such expectations from the perspective of the host-country population. Niemann et al. (2024), for example, study how working adults in Germany form expectations about migrants' occupational future and integration efforts depending on migrant characteristics such as origin, skill level, and reason for migration. Related research demonstrates that individuals also hold beliefs about discrimination. Haaland and Roth (2023) document substantial heterogeneity in beliefs about racial discrimination in hiring in the United States. Although their study concerns racial discrimination rather than migration, it demonstrates the conceptual distinction between beliefs about discrimination and actual discriminatory behavior.

Taken together, existing research provides evidence on several individual dimensions of migrants' expected integration, but evidence examining expectations across multiple economic and social outcomes remains limited. This thesis contributes to this literature by examining expectations concerning employment, income, discrimination, and social integration for Ukrainian and Syrian refugee migrants and economic migrants in Germany.

## **2.3 Differential Perceptions of Ukrainian and Syrian Migrants**

Earlier literature shows that responses to refugees in Germany already differed across groups and changed over time during and after the 2015–2016 refugee inflow. In particular, country of origin was associated with differences in acceptance of refugees, while other characteristics such as migration circumstances and family composition also shaped public responses (Bansak et al., 2016; Czymara and Schmidt-Catran, 2016, 2017; Plener et al., 2017; Liebe et al., 2018). Czymara and Schmidt-Catran (2017), for example, show that following the events of New Year's Eve 2015/2016, the acceptance of refugees generally increased, whereas the acceptance of immigrants of North African and Arab origin declined. Other research demonstrates that attitudes also varied with local exposure to the refugee inflow. Albarosa and Elsner (2023), for instance, document heterogeneous changes in attitudes across German regions depending on the size of the local refugee inflow. Looking at a later period, Przechomski et al. (2026) find that, after an initial increase in negative sentiment, attitudes gradually became more positive between 2017 and 2021.

The arrival of Ukrainian refugees following Russia's invasion of Ukraine in 2022 pro-

vided a new context for examining differences in responses to refugee groups in Europe. Existing research demonstrates that Ukrainian refugees have generally received more favorable public and institutional responses than Syrian refugees (Moise et al., 2024; Drewski and Gerhards, 2025; Das, 2025). Evidence from Germany similarly shows greater willingness to help Ukrainian refugees, more positive perceptions of their warmth and integration prospects, and greater solidarity toward them than toward Syrian refugees (Echterhoff et al., 2023; Schmidt et al., 2024; Stefaniak et al., 2025). These findings therefore suggest that public responses are shaped not only by refugee status itself but also by refugees' country of origin.

Several explanations have been proposed for these differences. Theories of group position and perceived group threat suggest that attitudes toward immigrant groups depend partly on how these groups are positioned and perceived relative to the host population. Building on the group-position perspective of Blumer (1958), Quillian (1995) argues that prejudice can reflect perceived collective threat and the relative position of the in-group and out-group. In the context of Ukrainian and Syrian migrants, existing studies similarly point to differences in perceived cultural similarity, threat, and deservingness (Echterhoff et al., 2023; Schmidt et al., 2024; Stefaniak et al., 2025). At the broader political level, Drewski and Gerhards (2025) show that Syrian and Ukrainian refugees were framed differently in Germany and Poland and link these differences to cultural and geopolitical factors that shaped state responses to the two groups.

These findings provide reason to expect that these differences between Ukrainian and Syrian migrants may also be reflected in beliefs about their integration outcomes. In line with this literature, I develop the following hypothesis:

H3a: Holding other migrant characteristics constant, respondents expect Ukrainian migrants to achieve more favorable labor-market and social integration outcomes than Syrian migrants.

In addition, the previous discussion in Section 2.2 provides reason to expect that individuals also form beliefs about the discrimination in labor-market outcomes of migrants. Combined with the less favorable perceptions of Syrian relative to Ukrainian migrants documented in this section, I developed the following hypothesis:

H3b: Holding other migrant characteristics constant, respondents expect Syrian migrants to face greater discrimination in the German labor market than Ukrainian migrants.

Further, I am interested in whether the difference between Ukrainian and Syrian migrants may depend on their reason for migration. Much of the literature above discussing differential responses toward the two groups concerns refugees specifically, suggesting that country of origin may be particularly important when migrants are evaluated in a refugee context. By contrast, evaluations of economic migrants may place relatively greater weight on characteristics related to their expected economic contribution and may not differ significantly by the country of origin. I therefore develop the following hypothesis:

H3c: Differences between Ukrainian and Syrian migrants in expected integration and discrimination outcomes are greater when migrants are presented as refugees than when they are presented as economic migrants.

In addition, I propose exploring whether differences between Ukrainian and Syrian migrants vary with their level of human capital, and in particular whether higher human capital reduces the country-of-origin gap in expected integration outcomes.

## **2.4 Belief Formation Through Contact, Experience, Information, and Memory**

Intergroup contact is one potential source from which individuals form perceptions of migrant groups. The contact hypothesis, originating with Allport (1954), proposes that contact between members of different groups can reduce prejudice under optimal contact conditions. A large subsequent literature generally finds an association between intergroup contact and lower prejudice, with Pettigrew and Tropp (2006) showing that these effects are not limited to situations in which all of Allport’s original conditions are satisfied. Evidence from migration and refugee contexts similarly shows that direct contact is associated with more favorable evaluations of migrants and refugees (Kotzur et al., 2019; De Coninck et al., 2021; Khalil and Naumann, 2022).

However, the amount of contact alone may not capture how intergroup experiences relate to perceptions. Contact can generate both positive and negative experiences, making it important to distinguish the frequency of contact from its valence. Positive contact is generally associated with more favorable attitudes toward migrants and refugees, whereas negative contact is associated with less favorable attitudes and behavioral tendencies (Laurence and Bentley, 2018; Kotzur et al., 2019; Bagci et al., 2023). De Coninck et al. (2021) similarly distinguish between the frequency and valence of intergroup contact and find that contact valence is particularly important for attitudes toward refugees. Moreover, increased exposure does not necessarily produce more favorable responses (Enos, 2019; Hangartner et al., 2019). Taken together, this literature suggests that the nature of respondents’ experiences with migrant groups may be more informative than exposure alone. I therefore develop the following hypothesis:

H4a: Conditional on contact frequency, more positive overall personal experiences with a migrant group are associated with more favorable beliefs about that group’s integration outcomes.

The importance of contact valence is consistent with a broader literature showing that personal social experiences can shape subsequent beliefs and behavior. Schwerter and Zimmermann (2020) experimentally show that positive and negative social experiences lead to substantial differences in subsequent willingness to trust, with beliefs about others’ trustworthiness playing an important role in these effects. However, the valence of contact

may also matter asymmetrically. In the intergroup-contact literature, Paolini et al. (2024) provide evidence that negative contact is more strongly associated with increased prejudice than positive contact is with reduced prejudice. Building on this evidence, I expect a similar asymmetry in the relationship between personal experiences with migrant groups and beliefs about their integration outcomes. This leads to the following hypothesis:

H4b: Negative personal experiences with a migrant group are more strongly associated with unfavorable beliefs about that group's integration outcomes than positive personal experiences are associated with favorable beliefs.

Personal experiences, however, are not the only source of information from which individuals may form beliefs about migrant groups. Experimental evidence shows that information about immigrants and refugees can influence subsequent perceptions and attitudes, although the direction and magnitude of these effects depend on the information provided and respondents' prior beliefs (Hayo and Neumeier, 2023; Dylong and Uebelmesser, 2026; Guenther, 2024). Media constitute a particularly important source of such information, providing indirect exposure to migrant groups even in the absence of direct personal contact (Schiappa et al., 2005; De Coninck et al., 2021). As with personal contact, the relationship between media exposure and attitudes may depend not only on the amount of exposure but also on its content and framing, as media representations of immigrants and refugees differ systematically in the economic, cultural, or security-related threats they emphasize (Atwell Seate and Mastro, 2016; Jacobs et al., 2016). Moreover, media exposure may itself be selective, as individuals may choose refugee-related media content based on their existing views (Lueders et al., 2019). Therefore, similar to the hypothesis about contact valence (H4a), I develop the following hypothesis:

H5: Conditional on media-exposure frequency, more positive overall media information concerning a migrant group is associated with more favorable beliefs about that group's integration outcomes.

Finally, past experiences and pieces of information may differ in how memorable they are for the respondent. Models of associative memory suggest that recall is selective and cue-dependent, such that experiences related to the current context are more likely to come to mind and influence beliefs (Bordalo et al., 2023; Enke et al., 2024). Consistent with this mechanism, Graeber et al. (2024) further show that stories are more readily recalled than statistical information and consequently remain more influential on beliefs over time. Applied to the migration context, beliefs about Ukrainian and Syrian migrants may therefore depend not only on respondents' overall exposure to each group, but also on the personal experiences and media information that are most memorable and on the valence of these memories. This leads to my last hypotheses:

H6a: Conditional on contact frequency and overall contact valence, the valence of respondents' most memorable personal experience with a migrant group is associated

with beliefs about that group’s integration outcomes.

H6b: Conditional on media-exposure frequency and overall media tone, the valence of respondents’ most memorable media information about a migrant group is associated with beliefs about that group’s integration outcomes.

### 3 Experimental Design

This thesis proposes a survey experiment to examine how Germans form beliefs about the labor-market and social integration outcomes of Ukrainian and Syrian migrants. The proposed study consists of a factorial vignette experiment, a set of outcome measures, and a respondent questionnaire. The following sections describe each part of the proposed experimental design and the survey procedure.

#### 3.1 Target Population and Proposed Sample

The target population consists of German citizens aged 18 or older who currently reside in Germany. Restricting eligibility to German citizens provides a clear operational definition of the German population whose beliefs about the integration of Ukrainian and Syrian migrants are the focus of this study. Similar citizenship-based sampling criteria have been used in previous studies examining attitudes toward refugees and migration in Germany (e.g., Hayo and Neumeier, 2023). Importantly, German citizenship is not used as a proxy for being native-born or having no migration background. Naturalized German citizens are therefore included in the target population, while respondents’ migration background is collected separately as a demographic characteristic in the respondent questionnaire. At the beginning of the survey, respondents would therefore be screened with respect to age, country of residence, and German citizenship. Respondents who do not satisfy these eligibility criteria would not proceed to the experimental part of the survey.

I propose using Qualtrics to program the survey, and the online panel provider Bilendi for the recruitment of participants. Bilendi is suitable for the present study because it provides access to a large and diverse pool of respondents in Germany and allows the use of demographic quotas to approximate the composition of the target population. The proposed sample would use quotas for age, gender, education, and region. Respondents would receive compensation for their participation according to the standard compensation provided by the panel, taking into account the expected duration of the survey.

The final sample size would be determined based on a power analysis of the proposed experimental design. The number of vignette evaluations per respondent, which also affects the required sample size, is discussed in Section 3.2.

### 3.2 Factorial Vignette Design

The proposed study employs a factorial vignette experiment in which respondents evaluate hypothetical migrants whose characteristics are independently varied across profiles. This approach is particularly suitable for the present research question because migrants differ simultaneously along several characteristics. By independently varying these characteristics, the design makes it possible to distinguish their effects on respondents' beliefs while holding the remaining characteristics constant. The proposed design builds on previous conjoint and factorial vignette experiments examining how migrant characteristics shape evaluations (Hainmueller and Hopkins, 2015; Bansak et al., 2016; Czymara and Schmidt-Catran, 2016).

The proposed vignettes describe migrants who have come to Germany and are currently looking for stable employment. Each profile varies independently along five binary dimensions: gender, country of origin, reason for migration, human capital, and German-language proficiency. These vignette attributes are selected to capture migrant characteristics that are central to the research questions and that previous research has shown to influence perceptions of migrants. Country of origin is varied between (1) Ukrainian and (2) Syrian migrants and represents the central comparison of the study. Reason for migration distinguishes between the two types of migrants: (1) refugees who fled the wars in Ukraine or Syria and (2) economic migrants who came to Germany in search of better employment opportunities. Educational attainment captures differences in migrants' human capital, which previous vignette and conjoint studies have shown to influence evaluations of immigrants (Hainmueller and Hiscox, 2010; Hainmueller and Hopkins, 2015; Czymara and Schmidt-Catran, 2016). The design distinguishes between (1) migrants without a formal vocational qualification and (2) migrants with a university degree. German-language proficiency is varied between (1) limited and (2) good German skills. Language proficiency may be relevant both for social integration and for labor-market outcomes, since knowledge of the host-country language represents an important form of human capital (Czymara and Schmidt-Catran, 2016). Finally, gender distinguishes between (1) male and (2) female migrants. Gender is included because previous experimental research suggests that migrant gender can affect how immigrants and refugees are perceived, including in the German context (Czymara and Schmidt-Catran, 2017; Gereke et al., 2020). Randomizing gender therefore allows the effects of the other vignette characteristics to be estimated across both male and female migrant profiles.

As discussed previously, existing research shows that religion and perceived cultural distance can influence attitudes toward immigrants (e.g., Czymara and Schmidt-Catran, 2016). However, in this study, religious affiliation is deliberately left unspecified. The main interest of this thesis is in the overall beliefs that respondents associate with Ukrainian and Syrian migrants. Explicitly including religion as an additional vignette attribute would not only make the design more complex, but could also change the beliefs that respondents would otherwise associate with the migrant's country of origin. For this reason, religion

is not varied separately in the experiment. The estimated country-of-origin effect should therefore be interpreted as an overall effect of Ukrainian versus Syrian origin, which may also reflect perceived cultural or religious differences between the two groups.

Table 1 summarizes the five vignette attributes and their respective levels and shows how the randomized characteristics relate to the hypotheses developed in Chapter 2.

Table 1: Vignette Attributes and Attribute Levels

| <b>Vignette attribute</b>   | <b>Level 1</b>                     | <b>Level 2</b>    | <b>Hypothesis</b> |
|-----------------------------|------------------------------------|-------------------|-------------------|
| Country of origin           | Syria                              | Ukraine           | H3a, H3b, H3c     |
| Reason for migration        | Economic migrant                   | War refugee       | H2, H3c           |
| Human capital               | No formal vocational qualification | University degree | H1a               |
| German-language proficiency | Limited                            | Good              | H1b               |
| Gender                      | Male                               | Female            | —                 |

*Notes:* The table presents the five independently randomized attributes used to construct the migrant profiles. Gender is included as an additional randomized characteristic but is not associated with a separate hypothesis.

The five binary attributes generate a full factorial design of  $2^5 = 32$  possible migrant profiles. The vignettes would be worded as follows:

A [male/female] from [Ukraine/Syria] came to Germany. [He/She] came to Germany as [a refugee after fleeing the war in his/her country / an economic migrant in search of better employment and economic opportunities in Germany]. [He/She] [has no formal vocational qualification / has completed a university degree], speaks German [only to a limited extent / well], and is currently looking for stable employment.

Since evaluating all possible profiles would place a substantial burden on respondents, I propose that each respondent evaluate a randomly selected subset of six vignettes. This provides several profile evaluations per respondent while keeping the survey manageable, particularly because each vignette is followed by several belief questions about migrant outcomes. The choice therefore reflects a trade-off between obtaining sufficient statistical information and limiting cognitive burden and respondent fatigue, which are important considerations in factorial survey designs (Sauer et al., 2011). The exact number of vignettes could be further assessed through pilot testing.

Prior to implementation, the survey would be pilot tested with approximately 50 respondents from the target population. The pilot would be used to assess survey completion time, attrition, potential respondent fatigue, and the performance of the attention and

data-quality checks. It would also be used to ensure that the vignette wording, questions, and response scales are understandable and that the Qualtrics randomization functions correctly. In particular, dropout patterns and response times across the survey would be examined to assess whether six vignette evaluations can reasonably be completed without imposing excessive burden. Based on the pilot results and respondent feedback, the exact number of vignettes and, where necessary, the wording or survey structure could be adjusted before the main study. The order in which the vignette profiles are presented to each respondent would also be randomized.

### 3.3 Outcome Variables

The five outcome questions would be answered right after each vignette. They are designed to capture respondents' beliefs about the labor-market, discrimination, and social-integration outcomes of the migrant described in each profile.

The outcomes cover three aspects: beliefs about labor-market integration outcomes (expected employment and expected income), beliefs about labor-market discrimination (expected hiring and workplace treatment), and beliefs about social integration. The exact wording of all outcome questions and response scales is provided in Appendix A.2.

The first two outcome questions capture beliefs about the migrant's labor-market integration. As discussed in Section 2.2, labor-market integration is reflected in several dimensions, including employment, earnings, and occupational outcomes (Brell et al., 2020; Fasani et al., 2024). Therefore, to capture the clearest and the most interpretable dimensions, the proposed experiment measures respondents' beliefs about labor-market integration based on expected employment and expected income. First, respondents state the probability that the migrant will find regular paid employment subject to social security contributions within the next two years. The type of employment in this question is formulated in a way to exclude informal, marginal, or unstable jobs. The timeframe of two years is chosen because it is long enough for job search, language learning, and recognition of qualifications of the migrants in Germany, but still short enough for respondents to imagine. Responses would be recorded on a scale from 0 to 100 percent. This probability scale provides a direct quantitative measure of respondents' beliefs about employment prospects. Second, respondents indicate the migrant's expected monthly net income five years from now by selecting one of seven income categories.

Two additional measures capture respondents' beliefs about discrimination in the German labor market. The first concerns hiring discrimination and asks respondents how likely is it that the migrant would be treated less favorably in the hiring process than an otherwise comparable German applicant. The second concerns discrimination after entering employment and measures the expected likelihood of unequal treatment in the workplace, for example in areas such as wages, promotion opportunities, or interactions with colleagues. Both outcomes are measured on seven-point scales, with higher values indicating greater agreement and therefore higher expected discrimination.

The final outcome captures beliefs about social integration. Respondents assess how likely the migrant is to successfully integrate socially in Germany within the next five years. To give respondents a common understanding of social integration, the question defines it as the migrants building regular social contacts with German citizens within the next five years. Responses are recorded on a seven-point scale, with higher values indicating more favorable beliefs about social integration.

Combined, these five measures capture different dimensions of respondents' beliefs about migrant integration rather than a single overall evaluation. The labor-market variables provide the main outcomes for H1a, H1b, H2, and H3a, while the discrimination measures provide the outcomes for H3b and the social-integration measure provides an additional outcome for H1b and H3a. All five outcome measures are used to examine H3c. In the empirical analysis, the outcome measures are analyzed separately rather than being combined into a single integration index.

### 3.4 Respondent Questionnaire

After answering all outcome questions for the vignette profiles, respondents would complete an additional questionnaire measuring personal contact experiences, media exposure, general beliefs, and their background characteristics. These questions are specifically asked after the vignette experiment to avoid influencing respondents' answers to the vignettes. For questions that are asked separately for Ukrainian and Syrian migrants, the order of the two groups would be randomized where feasible. The exact wording of all survey questions and their corresponding response options is provided in Appendix A.

#### 3.4.1 Personal Contact and Experience

**Contact frequency.** Respondents are asked separately about the frequency of workplace and non-work contact with Ukrainian and Syrian migrants during the past 12 months. Non-work contact refers to interactions outside of a workplace setting, for example as friends, neighbours, acquaintances, through clubs, or in other social settings. Both questions use the same seven-point frequency scale ranging from 1 (never) to 7 (several times per day). For more details on the exact wording of the questions and response scales, see Appendix A.3. For each migrant group, the two measures are combined into an overall contact exposure measure  $C_{ig}$ , which is constructed as the mean of the available workplace and non-work contact measures:

$$C_{ig} = \frac{C_{ig}^{\text{Work}} + C_{ig}^{\text{Non-Work}}}{2},$$

where  $C_{ig}^{\text{Work}}$  and  $C_{ig}^{\text{Non-Work}}$  denote workplace and non-work contact, respectively, for respondent  $i$  and migrant group  $g$ , where  $g \in \{\text{Ukrainian, Syrian}\}$ . Higher values indicate more frequent overall contact with the respective group. If a person is retired, unemployed, a student, or not currently working, they would answer that workplace contact is not

applicable for them and it would be marked as missing instead of reporting it as zero contact with the migrant group in the workplace. In this case, overall contact is based on non-work contact only,

$$C_{ig} = C_{ig}^{\text{NonWork}}.$$

This contact measure can be used in the specifications for H4–H6, because these hypotheses concern respondents’ overall contact experiences with the migrant group. Additionally, the two contact types can be analyzed separately to examine whether workplace and non-work contact are differently associated with respondents’ beliefs because contact in the workplace is not necessarily comparable to social contact with friends or neighbours.

**Overall contact quality.** If respondents report having had contact with Ukrainian or Syrian migrants, they are asked to evaluate the overall quality of their contact experiences with each group. The question is answered on a seven-point scale ranging from 1 (very negative) to 7 (very positive), with 4 representing a neutral experience that is neither positive nor negative. This measure is denoted by  $Q_{ig}$  and is used to examine H4a, conditional on respondents’ overall contact exposure  $C_{ig}$ .

To examine whether negative experiences are more strongly associated with respondents’ beliefs than positive experiences, as proposed in H4b, the overall experience measure is additionally separated into positive and negative components:

$$Positive_{ig} = \max(Q_{ig} - 4, 0) \quad \text{and} \quad Negative_{ig} = \max(4 - Q_{ig}, 0),$$

where  $Positive_{ig}$  captures the positivity of respondents’ overall contact experiences and  $Negative_{ig}$  captures the negativity of respondents’ overall contact experiences. Both of them are designed to range from 0 to 3 with higher values indicating more positive or negative experiences, respectively. Both take the value zero from a neutral response of  $Q_{ig} = 4$ . These separate measures make it possible to examine whether negative and positive experiences are differently associated with respondents’ beliefs.

**Most memorable experience.** If respondents report having had contact with Ukrainian or Syrian migrants, they are also asked to state whether a particular experience immediately comes to mind when thinking about their personal experiences. They would only respond with “Yes” or “No”.

**Valence of most memorable experience.** If their answer is “Yes”, they are asked to evaluate this most memorable experience, denoted by  $E_{ig}$ , on a seven-point scale. The valence of the most memorable experience is denoted by  $V_{ig}^E$  and is used to examine H6a while controlling for overall contact frequency and quality. For respondents who report having had contact with a migrant group but do not have a particular experience that comes to mind,  $V_{ig}^E$  is coded as missing.

### 3.4.2 Media Exposure and Information

**Media exposure.** Respondents are asked how often they come across information about Ukrainian and Syrian migrants living in Germany through media channels such as television, newspapers, online news, social media, radio, podcasts, etc. The question is asked separately for each migrant group using a seven-point scale ranging from 1 (never) to 7 (several times per day). The resulting measure of media-exposure frequency is denoted by  $M_{ig}$ , with higher values indicating more frequent exposure to media information.

**Overall media tone.** Respondents are then asked to evaluate the tone (valence) of the media information they have encountered about each migrant group using a seven-point scale ranging from 1 (very negative) to 7 (very positive), with 4 representing neutral or balanced information. The resulting measure,  $T_{ig}$ , captures the perceived tone of the media information. This measure is used to examine H5 while controlling for the frequency of media exposure  $M_{ig}$ .

**Most memorable media information.** If respondents have had media exposure, similar to the most memorable personal experience, respondents are asked whether a particular piece of media information immediately comes to mind. They would respond with either “Yes” or “No”. The most memorable piece of media information is denoted by  $I_{ig}$  which stands for information.

**Valence of most memorable media information.** Respondents who recall a particular piece of media information,  $I_{ig}$ , are asked to evaluate it on a seven-point scale. The valence of the most memorable media information is denoted by  $V_{ig}^I$ , with higher values indicating more positive recalled information. This measure is used to examine H6b while controlling for media-exposure frequency and overall media tone.

The measures of memorable personal experiences and media information differ from controlled experimental settings in which researchers manipulate the experiences participants encounter (Schwerter and Zimmermann, 2020) or observe the information participants receive and can subsequently assess recall against this information (Graeber et al., 2024). In the present thesis, both personal experiences and media information exposure occur naturally as positive or negative experiences in the lives of the respondents and are not observed by the researcher. The measures therefore capture respondents’ current subjective evaluations of the information that is most salient in memory rather than its objective valence or recall accuracy. This is relevant for the present thesis because the focus is on how respondents’ own perceptions of certain experiences and information are associated with their beliefs.

### 3.4.3 General Beliefs About Integration

The questionnaire also includes some questions on more general group-level beliefs about the integration of Ukrainian and Syrian migrants in Germany, capturing respondents’

broader perceptions of the two migrant groups. These measures are not used as primary outcomes for H1–H6. Instead, they can be used in descriptive and exploratory analyses that may help interpret mechanism behind why respondents’ beliefs differ between Ukrainians and Syrians.

**Integration over time.** Integration itself can also develop with time spent in the host country and can also differ across dimensions such as labor-market and social integration. Previous research shows that migrants’ labor-market integration tends to improve with duration of residence, including through higher employment and earnings, while initial labor-market disadvantages can narrow over time (Brücker et al., 2020; Fasani et al., 2022). There is also supporting evidence for social integration improving over time. Longer residence can be associated with greater interaction with the host-country population and stronger social integration (Niehues et al., 2021; Hannafi and Marouani, 2023). Such developments may also shape the beliefs of the general German population about whether migrants become better integrated the longer they live in Germany.

Therefore, the questionnaire is designed to measure these two beliefs. Respondents are asked whether they agree with the statements that (1) migrants become better integrated into the German labor market the longer they live in Germany, and that (2) they become more socially integrated the longer they live in Germany. Responses are evaluated on a seven-point scale ranging from 1 (strongly disagree) to 7 (strongly agree). The resulting measures are denoted by  $TimeLM_i$  and  $TimeSoc_i$ , capturing general beliefs that labor-market and social integration, respectively, improve with duration of residence.

**Establishment of Co-Ethnic Communities.** The potential role of co-ethnic networks may depend on how established migrant communities are perceived to be. Respondents are therefore asked whether they agree that the Ukrainian and Syrian communities in Germany are well established. Responses are recorded on a seven-point scale. The resulting measure is denoted by  $Comm_{ig}$ , with higher values indicating a stronger belief that the respective migrant community is well established in Germany.

**Community Networks: Labor-Market Facilitation.** Respondents may hold significant beliefs about the different integration outcomes associated with community networks. The first mechanism is that co-ethnic communities may facilitate migrants’ labor-market integration by providing access to information and personal networks that can help them find employment. Evidence from Germany shows that immigrants located in areas with larger co-ethnic networks are more likely to find employment shortly after arrival and are more likely to find their first job through personal contacts (Battisti et al., 2022). Respondents are therefore asked whether they agree that co-ethnic communities in Germany help newly arrived migrants find employment using a seven-point scale. The measure is denoted by  $FacilEmp_{ig}$ .

**Community Networks: Reduced Interaction with Germans.** The second mechanism is that co-ethnic communities may also affect social integration by reducing mi-

grants’ interaction with the host-country population. Evidence from Germany suggests that greater access to co-ethnic communities can reduce social interaction with natives and may also negatively affect the language ability of migrants (Danzer and Yaman, 2013, 2016). Respondents are therefore asked whether they agree with the statement that being able to rely on established Ukrainian and Syrian communities reduces the need for newly arrived migrants to interact regularly with Germans. Responses are recorded on a seven-point scale. The resulting measure is denoted by  $RedInt_{ig}$ .

#### 3.4.4 Respondent Controls

Lastly, the questionnaire collects respondent controls which are standard demographic and socioeconomic characteristics. These include respondents’ age, gender, education, employment status, household income, region of residence in Germany, migration background, religious affiliation, and political orientation. These variables are used to describe the sample and as control variables. The exact wording, response categories, and coding of these variables are provided in Appendix A.6.

### 3.5 Data Quality and Processing

Insufficient attention and careless responding can reduce the quality of data collected through self-administered online surveys, making procedures for identifying potentially low-quality responses important (Berinsky et al., 2014; Curran, 2016). The proposed survey would therefore include a simple instructed-response attention check and would record survey completion times. Available data-quality controls provided by Qualtrics and the Bilendi panel would additionally be used, where applicable, to identify duplicate or automated responses.

Respondents would be flagged for potential low-quality responding based on failure of the attention check or implausibly short completion times. The precise criteria for identifying such responses would be specified following pilot testing and prior to the main analysis. Rather than automatically excluding respondents based on a single indicator, the different indicators of response quality would be considered together. Robustness analyses could additionally examine whether the results are sensitive to the inclusion of respondents flagged for potential low-quality responding.

Data processing would distinguish between low-quality responses and legitimate missing or not-applicable responses. All displayed survey questions would require a response before respondents could proceed. Where appropriate, particularly for sensitive respondent characteristics, an explicit “Prefer not to say” option would be provided. Missing values may therefore arise from explicit nonresponse options, questions that are not applicable to a respondent, or conditional questions that are not displayed based on preceding answers. These values would not automatically result in respondent-level exclusion but would be treated according to the respective variable. The pilot study would also be used to assess the frequency of missing responses and whether the planned sample size provides

a sufficient number of observations for the corresponding analyses.

A detailed overview of the attention check, data-quality criteria, treatment and coding of missing and not-applicable responses, and construction of derived variables is provided in Appendix B.

## 4 Empirical Strategy

### 4.1 Experimental Analysis: Identification and Tests of H1–H3

The first part of the empirical analysis examines how the experimentally varied characteristics of the migrant profiles affect respondents' beliefs about migrants' labor-market and social integration outcomes. The unit of observation is a vignette evaluation, where respondent  $i$  evaluates migrant profile  $j$ . Since each respondent evaluates several migrant profiles, the analysis contains multiple observations for each respondent.

The factorial vignette design independently randomizes five characteristics across profiles: country of origin, reason for migration, gender, human capital, and German-language proficiency (see Chapter 3). Random assignment makes these characteristics independent of respondents' pre-existing characteristics and, in expectation, of the other randomized profile attributes. The experimental variation can therefore be used to identify the causal effect of each presented migrant characteristic on respondents' stated beliefs. In other words, the experiment identifies how respondents' beliefs change when one characteristic of the migrant profile changes. For example, it identifies how respondents' beliefs differ when an otherwise identical migrant is presented as Ukrainian rather than Syrian.

#### 4.1.1 Variables and Baseline Specification

**Baseline specification.** The five outcome variables introduced in Section 3.3 are analyzed separately. Let  $Y_{ij}^k$  denote respondent  $i$ 's belief about outcome  $k$  for vignette profile  $j$ , where

$$k \in \{\text{Emp, Inc, Hire, Work, Soc}\}.$$

$Y_{ij}^{\text{Emp}}$  denotes the expected probability of obtaining regular paid employment within two years,  $Y_{ij}^{\text{Inc}}$  denotes expected monthly net income,  $Y_{ij}^{\text{Hire}}$  and  $Y_{ij}^{\text{Work}}$  denote expected discrimination in hiring and in the workplace, respectively, and  $Y_{ij}^{\text{Soc}}$  denotes expected social integration.

The randomized vignette characteristics are represented by five binary variables.  $Ukr_j$  equals one if profile  $j$  describes a Ukrainian migrant and zero if the migrant is Syrian.  $Ref_j$  equals one if the migrant is presented as a war refugee and zero if the migrant is presented as an economic migrant.  $Fem_j$  equals one for a female migrant and zero for a male migrant.  $HighSkill_j$  equals one if the migrant has completed a university degree and zero if the migrant has no formal vocational qualification. Finally,  $German_j$  equals one if the migrant speaks German well and zero if the migrant speaks German only to a

limited extent. These definitions imply that the reference profile is a male Syrian economic migrant without a formal vocational qualification who speaks German only to a limited extent.

For each outcome  $k$ , the baseline specification is

$$Y_{ij}^k = \alpha^k + \beta_1^k Ukr_j + \beta_2^k Ref_j + \beta_3^k Fem_j + \beta_4^k HighSkill_j + \beta_5^k German_j + \varepsilon_{ij}^k. \quad (1)$$

The coefficients  $\beta_1^k$  to  $\beta_5^k$  capture the average causal effects of the respective randomized vignette characteristics on outcome  $k$ , relative to their reference categories and averaging over the randomized values of the remaining profile characteristics. The baseline specifications are estimated using OLS, and the interpretation of the coefficients depends on the respective outcome measure. For the employment outcome, which is measured on a 0–100 probability scale, coefficients are expressed in percentage points. For the income outcome, the seven ordered categories are coded from 1 to 7, so coefficients are expressed in income-category points rather than as specific differences in euros. For the remaining outcomes, which are measured on seven-point response scales, coefficients are expressed in scale points. Since the ordinal response categories cannot necessarily be assumed to be equally spaced, ordered logit models are additionally estimated as robustness checks to examine whether the conclusions are sensitive to treating these categories as approximately equally spaced in the OLS specifications. The term  $\varepsilon_{ij}^k$  denotes the error term and captures variation in respondent  $i$ 's evaluation of profile  $j$  for outcome  $k$  that is not explained by the variables included in the regression.

**Adjusted specification.** Because the vignette attributes are independently randomized, respondent-level controls are not required for identification of the coefficients in equation (1). The unadjusted specification therefore constitutes the main specification for estimating the experimental effects. As an adjusted specification, respondent background characteristics can additionally be included:

$$Y_{ij}^k = \alpha^k + \beta_1^k Ukr_j + \beta_2^k Ref_j + \beta_3^k Fem_j + \beta_4^k HighSkill_j + \beta_5^k German_j + \boldsymbol{\delta}^{k'} X_i + \varepsilon_{ij}^k. \quad (2)$$

where  $X_i$  is a vector of respondent-level controls including age, gender, education, employment status, household net income, migration background, region of residence in Germany, religion, and political self-placement, and is denoted by

$$X_i = (Age_i, Gender_i, Edu_i, Emp_i, Inc_i, Mig_i, East_i, Religion_i, Right_i)'$$

The inclusion of  $X_i$  does not provide the source of identification for the randomized vignette effects, but this adjusted specification can be used to assess the robustness and potentially improve the precision of the experimental estimates.

**Standard errors.** Since each respondent evaluates multiple vignette profiles, observa-

tions belonging to the same respondent cannot be treated as statistically independent. Standard errors are therefore clustered at the respondent level in all vignette-level specifications, allowing the regression residuals to be correlated across the profiles evaluated by the same respondent. This accounts for the repeated-evaluation structure of the proposed experiment.

**Multiple testing.** Since several hypotheses are examined across multiple outcomes, the analysis accounts for multiple testing using the Holm procedure. The adjustment is conducted separately within each pre-specified hypothesis family. For H1a and H2, the respective families include the employment and income outcomes; for H1b and H3a, employment, income, and social integration; for H3b, hiring and workplace discrimination; and for H3c, all five outcomes. For the associational hypotheses H4a, H4b, H5, H6a, and H6b, the respective families include employment, income, and social integration. Unadjusted estimates and 95% confidence intervals are reported alongside the Holm-adjusted  $p$ -values.

#### 4.1.2 Main Effects: H1a, H1b, H2, H3a, and H3b

Since several hypotheses include more than one outcome variable, equation (1) is estimated separately for each relevant outcome. For each hypothesis, the coefficient of interest and its predicted sign are specified below. The estimated coefficient shows the direction and size of the effect, while 95% confidence intervals are reported to indicate the uncertainty around the estimate. The estimated effect is reported and interpreted separately for each outcome. For example, if the results differ between employment and income, this difference is reported explicitly rather than combining the outcomes into a single measure of support for the hypothesis. Whether the results provide support for the respective hypothesis is determined from the pattern of the estimates, considering their predicted direction, magnitude, and statistical uncertainty. This avoids imposing a strict rule for hypothesis support before the empirical results are observed. Equation (2), which additionally includes respondent-level controls, is estimated as a robustness check.

**Testing H1a.** H1a predicts that migrants with higher human capital are expected to achieve more favorable labor-market integration outcomes. Equation (1) is therefore estimated for the employment and income outcomes,  $Y_{ij}^{\text{Emp}}$  and  $Y_{ij}^{\text{Inc}}$ . The coefficient of interest is  $\beta_4^k$  on  $HighSkill_j$ , which captures the causal effect of presenting a migrant with a university degree rather than with no formal vocational qualification on respondents' beliefs about the respective outcome. H1a predicts

$$\beta_4^{\text{Emp}} > 0 \quad \text{and} \quad \beta_4^{\text{Inc}} > 0.$$

Thus, positive coefficients indicate more favorable beliefs about the respective labor-market outcomes for profiles with higher human capital.

**Testing H1b.** H1b predicts that migrants with higher German-language proficiency are expected to achieve more favorable labor-market and social-integration outcomes. Equation (1) is therefore estimated  $Y_{ij}^{\text{Emp}}$ ,  $Y_{ij}^{\text{Inc}}$ , and expected social integration  $Y_{ij}^{\text{Soc}}$ . The coefficient of interest is  $\beta_5^k$  on  $\text{German}_j$ , which captures the causal effect of presenting a migrant as speaking German well rather than only to a limited extent on respondents' beliefs. H1b predicts

$$\beta_5^{\text{Emp}} > 0, \quad \beta_5^{\text{Inc}} > 0, \quad \text{and} \quad \beta_5^{\text{Soc}} > 0.$$

Thus, positive coefficients indicate more favorable beliefs about the respective outcomes for profiles with higher German-language proficiency.

**Testing H2.** H2 predicts that refugees are expected to achieve less favorable labor-market integration outcomes than otherwise comparable economic migrants. Equation (1) is estimated for  $Y_{ij}^{\text{Emp}}$  and  $Y_{ij}^{\text{Inc}}$ . The coefficient of interest is  $\beta_2^k$  on  $\text{Ref}_j$ , which captures the causal effect of presenting a migrant as a war refugee rather than as an economic migrant on respondents' beliefs. H2 predicts

$$\beta_2^{\text{Emp}} < 0 \quad \text{and} \quad \beta_2^{\text{Inc}} < 0.$$

Thus, negative coefficients indicate less favorable beliefs about the respective outcomes for profiles presented as refugees rather than economic migrants.

**Testing H3a.** H3a predicts that Ukrainian migrants are expected to achieve more favorable labor-market and social-integration outcomes than otherwise comparable Syrian migrants. Equation (1) is estimated for  $Y_{ij}^{\text{Emp}}$ ,  $Y_{ij}^{\text{Inc}}$ , and  $Y_{ij}^{\text{Soc}}$ . The coefficient of interest is  $\beta_1^k$  on  $\text{Ukr}_j$ , which captures the causal effect of presenting a migrant as Ukrainian rather than Syrian on respondents' beliefs. H3a predicts

$$\beta_1^{\text{Emp}} > 0, \quad \beta_1^{\text{Inc}} > 0, \quad \text{and} \quad \beta_1^{\text{Soc}} > 0.$$

Thus, positive coefficients indicate more favorable beliefs about the respective outcomes for profiles presented as Ukrainian rather than Syrian.

**Testing H3b.** H3b examines country-of-origin differences in expected labor-market discrimination and predicts that Syrian migrants are expected to face greater discrimination than otherwise comparable Ukrainian migrants. The hypothesis is tested using the country-of-origin indicator,  $\text{Ukr}_j$ , with  $Y_{ij}^{\text{Hire}}$  and  $Y_{ij}^{\text{Work}}$  as the dependent variables. An important difference from the preceding outcomes is that higher values of both discrimination measures indicate a less favorable outcome, namely greater expected discrimination. Since Syria is the reference category and  $\text{Ukr}_j = 1$  represents Ukraine, H3b therefore predicts

$$\beta_1^{\text{Hire}} < 0 \quad \text{and} \quad \beta_1^{\text{Work}} < 0.$$

Negative coefficients thus indicate that profiles presented as Ukrainian are expected to face less hiring and workplace discrimination than otherwise comparable Syrian profiles.

#### 4.1.3 Interaction Effects: H3c and Additional Analysis

**Testing H3c.** While H3a and H3b examine average country-of-origin differences in expected integration and discrimination outcomes, H3c examines whether the size of these differences varies systematically between the two reasons for migration. Specifically, H3c predicts that differences between Ukrainian and Syrian migrants are greater when migrants are presented as refugees than when they are presented as economic migrants.

To test this hypothesis, an interaction term between country of origin and reason for migration is added to the baseline specification:

$$Y_{ij}^k = \alpha^k + \beta_1^k Ukr_j + \beta_2^k Ref_j + \beta_3^k Fem_j + \beta_4^k HighSkill_j + \beta_5^k German_j + \beta_6^k (Ukr_j \times Ref_j) + \varepsilon_{ij}^k. \quad (3)$$

Since  $Ref_j = 0$  represents economic migrants,  $\beta_1^k$  captures the Ukrainian–Syrian difference among economic migrants. When  $Ref_j = 1$ , the corresponding Ukrainian–Syrian difference among refugees is given by  $\beta_1^k + \beta_6^k$ . Thus,  $\beta_6^k$  captures how the country-of-origin difference changes when migrants are presented as refugees rather than economic migrants. For employment, income, and social integration, higher values represent more favorable outcomes and H3c therefore predicts

$$\beta_6^{Emp} > 0, \quad \beta_6^{Inc} > 0, \quad \beta_6^{Soc} > 0.$$

For hiring and workplace discrimination, higher values instead represent greater expected discrimination. Since Ukrainian migrants are expected to face less discrimination, a larger country-of-origin difference among refugees corresponds to a more negative Ukrainian–Syrian difference. H3c therefore predicts

$$\beta_6^{Hire} < 0 \quad \text{and} \quad \beta_6^{Work} < 0.$$

In addition to the interaction coefficient, the conditional country-of-origin difference among refugees,  $\beta_1^k + \beta_6^k$ , is reported together with its 95% confidence interval. This shows whether the Ukrainian–Syrian difference among refugees itself is in the direction predicted by H3a and H3b, while  $\beta_6^k$  tests whether this difference is greater among refugees than among economic migrants.

**Additional human-capital analysis.** In addition to H3c, the analysis similarly explores whether the Ukrainian–Syrian difference in expected integration outcomes varies with migrants’ human capital. For this purpose, the equation (1) is extended separately by the interaction  $Ukr_j \times HighSkill_j$ . If higher human capital reduces the Ukrainian–Syrian gap in expected integration outcomes, the coefficient on this interaction is expected to be

negative. This analysis is treated as supporting rather than as a separate hypothesis.

## 4.2 Associational Analysis: Contact, Experience, Media, and Memory

This section examines H4–H6, which concern the relationship between respondents’ personal contact, media exposure, most memorable experiences and information and their beliefs about migrant integration. Unlike the vignette characteristics analyzed in Section 4.1, these respondent-level measures are not randomly assigned. The corresponding coefficients are therefore interpreted as conditional associations rather than causal effects. The analysis controls for relevant observed characteristics. These specifications can reduce some sources of confounding but cannot rule out selection, reverse causality, or unobserved factors that may jointly influence respondents’ experiences, information exposure, and beliefs. These identification limitations and possible approaches for addressing them in future research are discussed further in Section 5. As in the experimental analysis, standard errors are clustered at the respondent level to account for repeated vignette evaluations.

### 4.2.1 Contact and Experience: H4a–H4b

**Testing H4a.** H4a predicts that, conditional on contact frequency, more positive overall personal experiences with a migrant group are associated with more favorable beliefs about its integration outcomes.

Because contact and experience are measured separately for Ukrainian and Syrian migrants, these measures are matched to the country of origin presented in each vignette. Let  $g(j) \in \{\text{Ukrainian}, \text{Syrian}\}$  denote the country-of-origin group presented in vignette  $j$ . Thus,  $C_{i,g(j)}$  and  $Q_{i,g(j)}$  refer to respondent  $i$ ’s contact frequency and overall contact quality, respectively, with the group presented in vignette  $j$ . For example, a Ukrainian vignette is matched to the respondent’s reported contact and experience with Ukrainian migrants.

To test H4a, the following specification is estimated:

$$Y_{ij}^k = \alpha^k + \beta^{k'} Z_j + \gamma_C^k C_{i,g(j)} + \gamma_Q^k Q_{i,g(j)} + \delta^{k'} X_i + \varepsilon_{ij}^k, \quad (4)$$

where  $Z_j$  is the vector of randomized vignette characteristics,

$$Z_j = (\text{Ukr}_j, \text{Ref}_j, \text{Fem}_j, \text{HighSkill}_j, \text{German}_j)'$$

$X_i$  contains the respondent-level controls defined above. The coefficient of interest for H4a is  $\gamma_Q^k$ , which captures the association between the overall quality of contact with the relevant migrant group and respondent beliefs.  $C_{i,g(j)}$  is included so that the association between overall contact quality and beliefs is estimated conditional on how frequently

respondents have contact with the respective migrant group. Its coefficient,  $\gamma_C^k$ , captures the association between contact frequency and beliefs, holding overall contact quality and the other included variables constant.

H4a predicts that more positive contact is associated with more favorable beliefs about employment, income, and social-integration outcomes:

$$\gamma_Q^{\text{Emp}} > 0, \quad \gamma_Q^{\text{Inc}} > 0, \quad \gamma_Q^{\text{Soc}} > 0.$$

**Testing H4b.** H4b examines whether negative personal experiences are more strongly associated with respondents' beliefs than positive personal experiences. Using the positive and negative experience measures  $Positive_{ig}$  and  $Negative_{ig}$  constructed from  $Q_{ig}$  as described in Section 3.4.1, the two measures are included simultaneously in the following specification:

$$Y_{ij}^k = \alpha^k + \beta^{k'} Z_j + \gamma_C^k C_{i,g(j)} + \gamma_P^k Positive_{i,g(j)} + \gamma_N^k Negative_{i,g(j)} + \delta^{k'} X_i + \varepsilon_{ij}^k. \quad (5)$$

For the employment, income, and social-integration outcomes, more positive experiences are expected to be associated with more favorable beliefs, whereas more negative experiences are expected to be associated with less favorable beliefs. Thus,

$$\gamma_P^{\text{Emp}}, \gamma_P^{\text{Inc}}, \gamma_P^{\text{Soc}} > 0, \quad \gamma_N^{\text{Emp}}, \gamma_N^{\text{Inc}}, \gamma_N^{\text{Soc}} < 0.$$

H4b, however, predicts not only opposite directions but also a stronger association for negative than for equally strong positive experiences. This prediction can be tested by comparing the absolute magnitudes of their coefficients. For each relevant outcome  $k$ , H4b predicts

$$-\gamma_N^k > \gamma_P^k,$$

or equivalently,

$$\gamma_P^k + \gamma_N^k < 0.$$

The comparison therefore tests whether moving one unit toward a more negative experience is associated with a larger change in beliefs than an equivalent one-unit movement toward a more positive experience.

#### 4.2.2 Media Exposure and Tone: H5

**Testing H5.** H5 examines whether the overall valence of media information concerning a migrant group is associated with respondents' beliefs about that group's integration outcomes, conditional on the frequency of media exposure. As in the contact analysis, the group-specific media measures are matched to the country of origin presented in each

vignette. The following specification is estimated:

$$Y_{ij}^k = \alpha^k + \beta^{k'} Z_j + \gamma_M^k M_{i,g(j)} + \gamma_T^k T_{i,g(j)} + \delta^{k'} X_i + \varepsilon_{ij}^k. \quad (6)$$

The coefficient of interest is  $\gamma_T^k$  on  $T_{i,g(j)}$ , which captures the association between the overall tone of media information and respondent beliefs.  $M_{i,g(j)}$  is included so that the association between media tone and beliefs is estimated conditional on how frequently respondents encounter media information about the respective migrant group. Its coefficient,  $\gamma_M^k$ , captures the association between media-exposure frequency and beliefs, holding media tone and the other included variables constant.

For the employment, income, and social-integration outcomes, H5 predicts that more positive media information is associated with more favorable beliefs:

$$\gamma_T^{\text{Emp}} > 0, \quad \gamma_T^{\text{Inc}} > 0, \quad \gamma_T^{\text{Soc}} > 0.$$

#### 4.2.3 Memorable Experiences and Information: H6a–H6b

**Testing H6a.** H6a examines whether the valence of a respondent's most memorable personal experience with a migrant group is associated with beliefs, conditional on contact frequency and overall contact quality. The valence of the most memorable experience,  $V_{ig}^E$ , is observed only when the respondent reports that such an experience comes to mind ( $E_{ig} = 1$ ). The following specification is therefore estimated only for the observations satisfying  $E_{i,g(j)} = 1$ , and for each of the three integration outcomes,  $k \in \{\text{Emp}, \text{Inc}, \text{Soc}\}$ :

$$Y_{ij}^k = \alpha^k + \beta^{k'} Z_j + \gamma_C^k C_{i,g(j)} + \gamma_Q^k Q_{i,g(j)} + \gamma_E^k V_{i,g(j)}^E + \delta^{k'} X_i + \varepsilon_{ij}^k. \quad (7)$$

The coefficient of interest is  $\gamma_E^k$ . Thus, H6a examines whether memorable-experience valence is associated with beliefs even after respondents' overall experiences with the respective migrant group are taken into account. The direction, magnitude, and statistical uncertainty of  $\gamma_E^k$  are reported for each relevant outcome.

**Testing H6b.** H6b examines the corresponding relationship for respondents' most memorable media information. Similarly,  $V_{ig}^I$  is observed only when the respondent reports that a particularly memorable piece of media information exists,  $I_{ig} = 1$ . Then, the following specification is estimated for each of the three integration outcomes,  $k \in \{\text{Emp}, \text{Inc}, \text{Soc}\}$ :

$$Y_{ij}^k = \alpha^k + \beta^{k'} Z_j + \gamma_M^k M_{i,g(j)} + \gamma_T^k T_{i,g(j)} + \gamma_I^k V_{i,g(j)}^I + \delta^{k'} X_i + \varepsilon_{ij}^k. \quad (8)$$

The coefficient of interest is  $\gamma_I^k$ . H6b therefore examines whether the valence of memorable media information is associated with beliefs conditional on respondents' overall assessment of the media information they encounter. The direction, magnitude, and statistical uncertainty of  $\gamma_I^k$  are reported for each relevant outcome.

**Supporting analyses.** In addition to H6a and H6b, I propose comparing the coefficients of the overall and memorable measures to examine whether their associations with respondents’ beliefs differ. This suggests comparing  $\gamma_Q^k$  and  $\gamma_E^k$  for contact experiences, and  $\gamma_T^k$  and  $\gamma_I^k$  for media information. Since the measures are recorded on the same seven-point scales, this makes it possible to test whether the corresponding coefficients differ from one another. The comparisons are interpreted cautiously because respondents’ overall assessments may themselves partly reflect their most memorable experiences or information. Conversely, the overall and memorable measures may also differ substantially, for example when a respondent reports generally positive experiences but recalls a particularly negative experience as most memorable. In such cases, the analysis can show whether beliefs are more strongly associated with the overall or the memorable measure.

As an additional exploratory analysis, the specifications used to examine H4–H6 are also estimated using the two discrimination outcomes,  $Y_{ij}^{\text{Hire}}$  and  $Y_{ij}^{\text{Work}}$ . While H4–H6 focus on beliefs about integration outcomes, this extension examines whether respondents’ contact experiences and media information are also associated with beliefs about the discrimination faced by the respective migrant group.

### 4.3 General Beliefs and Exploratory Analyses

The measures of respondents’ more general beliefs introduced in Section 3.4.3 are analyzed exploratorily and provide complementary evidence on how respondents perceive differences between the two migrant groups.

Let  $B_{ig}$  denote one of the three group-specific belief measures which are collected separately for Ukrainian and Syrian migrants,

$$B_{ig} \in \{Comm_{ig}, FacilEmp_{ig}, RedInt_{ig}\}.$$

The following specification is estimated separately for each measure:

$$B_{ig} = \alpha_i + \theta Ukr_g + \varepsilon_{ig}, \tag{9}$$

where  $Ukr_g = 1$  when the evaluated group is Ukrainian and  $Ukr_g = 0$  when it is Syrian. The respondent fixed effect,  $\alpha_i$ , accounts for all characteristics that are constant within respondent across the two group evaluations. The coefficient of interest,  $\theta$ , therefore captures the average within-respondent difference between beliefs about Ukrainian and Syrian migrants for the respective measure.  $\theta > 0$  indicates a higher value for Ukrainian than for Syrian migrants, while  $\theta < 0$  indicates a higher value for Syrian than for Ukrainian migrants. The coefficient is interpreted as a difference in reported beliefs rather than as a causal effect of country of origin.

I next examine whether these group-specific general beliefs are associated with respondents’ evaluations of the vignette profiles. As in Section 4.2, each belief measure is matched to the country of origin presented in vignette  $j$ . The following specification is

estimated for each of the three integration outcomes,  $k \in \{\text{Emp, Inc, Soc}\}$ :

$$Y_{ij}^k = \alpha^k + \beta^{k'} Z_j + \lambda_B^k B_{i,g(j)} + \delta^{k'} X_i + \varepsilon_{ij}^k. \quad (10)$$

The coefficient of interest,  $\lambda_B^k$ , captures the conditional association between the respective group-specific general belief and  $Y_{ij}^k$ , holding the other independent variables constant. The specification is estimated separately for the three belief measures. For  $Comm_{i,g(j)}$ ,  $\lambda_B^k$  indicates how beliefs about how well established the respective migrant community is are associated with the vignette outcome. For  $FacilEmp_{i,g(j)}$ , it captures the association with beliefs that the respective community facilitates employment, and for  $RedInt_{i,g(j)}$ , it captures the association with beliefs that reliance on the respective community reduces interaction with Germans.

These exploratory analyses can help examine how respondents' broader beliefs about migrant communities are associated with their beliefs about migrants' integration outcomes and may thereby provide additional information for interpreting the mechanisms underlying these beliefs.

The variables  $TimeLM_i$  and  $TimeSoc_i$  capture a different dimension of respondents' general integration beliefs. They measure general beliefs about whether migrants become better integrated with time spent in Germany.  $TimeLM_i$  is related to the employment and income outcomes using the following specification:

$$Y_{ij}^k = \alpha^k + \beta^{k'} Z_j + \lambda_{TimeLM}^k TimeLM_i + \delta^{k'} X_i + \varepsilon_{ij}^k, \quad k \in \{\text{Emp, Inc}\}. \quad (11)$$

The coefficient  $\lambda_{TimeLM}^k$  captures the conditional association between stronger beliefs that migrants' labor-market integration improves with time and respondents' beliefs about the respective outcome. The corresponding relationship for social integration is estimated as

$$Y_{ij}^{\text{Soc}} = \alpha^{\text{Soc}} + \beta^{\text{Soc}'} Z_j + \lambda_{TimeSoc}^{\text{Soc}} TimeSoc_i + \delta^{\text{Soc}'} X_i + \varepsilon_{ij}^{\text{Soc}}. \quad (12)$$

Here,  $\lambda_{TimeSoc}^{\text{Soc}}$  captures the conditional association between stronger beliefs that migrants become more socially integrated with time and respondents' beliefs about the social integration of the vignette profile.

These specifications provide an exploratory test of whether the two types of beliefs are systematically related. For instance, respondents who more strongly believe that migrants become better integrated with time may also report more favorable beliefs about their future labor-market or social integration outcomes. However, because the vignette does not specify how long the migrant has already lived in Germany, respondents may perceive the migrant as relatively newly arrived, in which case their general beliefs about integration over time may be less directly reflected in their evaluation of the profile.

Finally, the five general-belief measures examined in this section may themselves be related. For example, beliefs about how well established migrant communities are may be associated with beliefs about whether these communities facilitate employment or reduce

interaction with Germans, and may also be related to respondents' more general beliefs about how integration develops over time. Examining these relationships may therefore provide additional information about how different beliefs about migrant integration are connected. Possible approaches for examining them more explicitly are discussed further in Section 5.

## **5 Limitations and Extensions**

### **5.1 Extensions**

### **5.2 Limitations and Identification Constraints**

## **6 Conclusion**

## References

- Albarosa, E. and Elsner, B. (2023). Forced migration and social cohesion: Evidence from the 2015/16 mass inflow in germany. *World Development*, 167:106228.
- Alesina, A., Miano, A., and Stantcheva, S. (2023). Immigration and redistribution. *The Review of Economic Studies*, 90(1):1–39.
- Allport, G. W. (1954). The nature of prejudice. *Reading/Addison-Wesley*.
- Almeida, S., Fernando, M., Hannif, Z., and Dharmage, S. C. (2015). Fitting the mould: The role of employer perceptions in immigrant recruitment decision-making. *The International Journal of Human Resource Management*, 26(22):2811–2832.
- Atwell Seate, A. and Mastro, D. (2016). Media’s influence on immigration attitudes: An intergroup threat theory approach. *Communication Monographs*, 83(2):194–213.
- Bagci, S. C., Baysu, G., Tercan, M., and Turnuklu, A. (2023). Dealing with increasing negativity toward refugees: A latent growth curve study of positive and negative intergroup contact and approach-avoidance tendencies. *Personality and Social Psychology Bulletin*, 49(10):1466–1478.
- Bansak, K., Hainmueller, J., and Hangartner, D. (2016). How economic, humanitarian, and religious concerns shape european attitudes toward asylum seekers. *Science*, 354(6309):217–222.
- Battisti, M., Peri, G., and Romiti, A. (2022). Dynamic effects of co-ethnic networks on immigrants’ economic success. *The Economic Journal*, 132(641):58–88.
- Berinsky, A. J., Margolis, M. F., and Sances, M. W. (2014). Separating the shirkers from the workers? making sure respondents pay attention on self-administered surveys. *American journal of political science*, 58(3):739–753.
- Blinder, S. (2015). Imagined immigration: The impact of different meanings of ‘immigrants’ in public opinion and policy debates in britain. *Political Studies*, 63(1):80–100.
- Blumer, H. (1958). Race prejudice as a sense of group position. *Pacific sociological review*, 1(1):3–7.
- Bordalo, P., Conlon, J. J., Gennaioli, N., Kwon, S. Y., and Shleifer, A. (2023). Memory and probability. *The Quarterly Journal of Economics*, 138(1):265–311.
- Brell, C., Dustmann, C., and Preston, I. (2020). The labor market integration of refugee migrants in high-income countries. *Journal of Economic Perspectives*, 34(1):94–121.
- Brücker, H., Kosyakova, Y., and Schuß, E. (2020). Fünf jahre seit der fluchtmigration 2015: Integration in arbeitsmarkt und bildungssystem macht weitere fortschritte. Technical report, IAB-Kurzbericht.

- Burns, P. and Gimpel, J. G. (2000). Economic insecurity, prejudicial stereotypes, and public opinion on immigration policy. *Political science quarterly*, 115(2):201–225.
- Curran, P. G. (2016). Methods for the detection of carelessly invalid responses in survey data. *Journal of Experimental Social Psychology*, 66:4–19.
- Czymara, C. S. and Schmidt-Catran, A. W. (2016). Wer ist in deutschland willkommen? *KZfSS Kölner Zeitschrift für Soziologie und Sozialpsychologie*, 68(2):193–227.
- Czymara, C. S. and Schmidt-Catran, A. W. (2017). Refugees unwelcome? changes in the public acceptance of immigrants and refugees in germany in the course of europe’s ‘immigration crisis’. *European Sociological Review*, 33(6):735–751.
- Danzer, A. M. and Yaman, F. (2013). Do ethnic enclaves impede immigrants’ integration? evidence from a quasi-experimental social-interaction approach. *Review of International Economics*, 21(2):311–325.
- Danzer, A. M. and Yaman, F. (2016). Ethnic concentration and language fluency of immigrants: Evidence from the guest-worker placement in germany. *Journal of Economic Behavior & Organization*, 131:151–165.
- Das, R. (2025). Changing contours of dealing with refugee crisis in eu: A comparative study of ukrainian and syrian refugee crisis. *Journal of Migration and Political Studies*, 3(1):42–56.
- De Coninck, D., Rodríguez-de Dios, I., and d’Haenens, L. (2021). The contact hypothesis during the european refugee crisis: Relating quality and quantity of (in) direct inter-group contact to attitudes towards refugees. *Group Processes & Intergroup Relations*, 24(6):881–901.
- Dietz, J., Joshi, C., Esses, V. M., Hamilton, L. K., and Gabarrot, F. (2015). The skill paradox: Explaining and reducing employment discrimination against skilled immigrants. *The International Journal of Human Resource Management*, 26(10):1318–1334.
- Drewski, D. and Gerhards, J. (2025). Why do states discriminate between refugee groups? understanding how syrian and ukrainian refugees were framed in germany and poland: D. drewski and j. gerhards. *American Journal of Cultural Sociology*, 13(2):271–297.
- Dustmann, C. and Preston, I. (2007). Racial and economic factors in attitudes to immigration. *The BE Journal of Economic Analysis & Policy*, 7(1).
- Dylong, P. and Uebelmesser, S. (2026). Intergroup contact and exposure to information about immigrants: Experimental evidence. *European Journal of Political Economy*, page 102883.

- Echterhoff, G., Thölmann, C., Gemein, R., and Knausenberger, J. (2023). Geflüchtete erster und zweiter klasse? reaktionen auf syrische und ukrainische geflüchtete in deutschland.
- Enke, B., Schwerter, F., and Zimmermann, F. (2024). Associative memory, beliefs and market interactions. *Journal of Financial Economics*, 157:103853.
- Enos, R. D. (2019). "causal effect of intergroup contact on exclusionary attitudes": Correction.
- Farashah, A., Blomquist, T., Al Ariss, A., and Guo, G. C. (2023). Perceived employability of skilled migrants: a systematic review and future research agenda. *The International Journal of Human Resource Management*, 34(3):478–528.
- Fasani, F., Frattini, T., and Minale, L. (2022). (the struggle for) refugee integration into the labour market: Evidence from europe. *Journal of Economic Geography*, 22(2):351–393.
- Fasani, F., Frattini, T., and Pirot, M. (2024). From refugees to citizens: Labor market returns to naturalization. *Centro Studi Luca d’Agliano Development Studies Working Paper*, (489).
- Gereke, J., Schaub, M., and Baldassarri, D. (2020). Gendered discrimination against immigrants: experimental evidence. *Frontiers in sociology*, 5:539303.
- Graeber, T., Roth, C., and Zimmermann, F. (2024). Stories, statistics, and memory. *The Quarterly Journal of Economics*, 139(4):2181–2225.
- Guenther, L. (2024). Revisiting immigration attitudes: How asylum seekers nuance previous findings.
- Haaland, I. and Roth, C. (2023). Beliefs about racial discrimination and support for pro-black policies. *Review of Economics and Statistics*, 105(1):40–53.
- Hainmueller, J. and Hiscox, M. J. (2010). Attitudes toward highly skilled and low-skilled immigration: Evidence from a survey experiment. *American political science review*, 104(1):61–84.
- Hainmueller, J. and Hopkins, D. J. (2014). Public attitudes toward immigration. *Annual review of political science*, 17(1):225–249.
- Hainmueller, J. and Hopkins, D. J. (2015). The hidden american immigration consensus: A conjoint analysis of attitudes toward immigrants. *American journal of political science*, 59(3):529–548.

- Hangartner, D., Dinas, E., Marbach, M., Matakos, K., and Xefferis, D. (2019). Does exposure to the refugee crisis make natives more hostile? *American political science review*, 113(2):442–455.
- Hannafi, C. and Marouani, M. A. (2023). Social integration of syrian refugees and their intention to stay in germany: C. hannafi, ma marouani. *Journal of Population Economics*, 36(2):581–607.
- Hayo, B. and Neumeier, F. (2023). Between fearmongers and samaritans: Does information provision affect attitudes towards the right of asylum in germany? *Kyklos*, 76(4):749–777.
- Jacobs, L., Meeusen, C., and d’Haenens, L. (2016). News coverage and attitudes on immigration: Public and commercial television news compared. *European Journal of Communication*, 31(6):642–660.
- Khalil, S. and Naumann, E. (2022). Does contact with foreigners reduce worries about immigration? a longitudinal analysis in germany. *European Sociological Review*, 38(2):189–201.
- Kotzur, P. F., Schäfer, S. J., and Wagner, U. (2019). Meeting a nice asylum seeker: Intergroup contact changes stereotype content perceptions and associated emotional prejudices, and encourages solidarity-based collective action intentions. *British Journal of Social Psychology*, 58(3):668–690.
- Laurence, J. and Bentley, L. (2018). Countervailing contact: Community ethnic diversity, anti-immigrant attitudes and mediating pathways of positive and negative inter-ethnic contact in european societies. *Social science research*, 69:83–110.
- Liebe, U., Meyerhoff, J., Kroesen, M., Chorus, C., and Glenk, K. (2018). From welcome culture to welcome limits? uncovering preference changes over time for sheltering refugees in germany. *PloS one*, 13(8):e0199923.
- Lueders, A., Prentice, M., and Jonas, E. (2019). Refugees in the media: Exploring a vicious cycle of frustrated psychological needs, selective exposure, and hostile intergroup attitudes. *European Journal of Social Psychology*, 49(7):1471–1479.
- Mayda, A. M. (2006). Who is against immigration? a cross-country investigation of individual attitudes toward immigrants. *The review of Economics and Statistics*, 88(3):510–530.
- Moise, A. D., Dennison, J., and Kriesi, H. (2024). European attitudes to refugees after the russian invasion of ukraine. *West european politics*, 47(2):356–381.
- Niehues, W., Rother, N., and Siegert, M. (2021). Spracherwerb und soziale kontakte schreiben bei geflüchteten voran: vierte welle der iab-bamf-soep-befragung von geflüchteten.

- Niemann, L., Naudascher, H., Kuyumcu, U., and Hertel, G. (2024). “how long are you gonna stay?” implications of residents’ perceptions of refugees’ occupational future time perspective in receiving countries: Niemann l. et al. *Journal of International Migration and Integration*, 25(3):1423–1443.
- Paolini, S., Gibbs, M., Sales, B., Anderson, D., and McIntyre, K. (2024). Negativity bias in intergroup contact: Meta-analytical evidence that bad is stronger than good, especially when people have the opportunity and motivation to opt out of contact. *Psychological bulletin*, 150(8):921.
- Pettigrew, T. F. and Tropp, L. R. (2006). A meta-analytic test of intergroup contact theory. *Journal of personality and social psychology*, 90(5):751.
- Plener, P. L., Groschwitz, R. C., Brähler, E., Sukale, T., and Fegert, J. M. (2017). Unaccompanied refugee minors in germany: attitudes of the general population towards a vulnerable group. *European child & adolescent psychiatry*, 26(6):733–742.
- Przechomski, L., Seddig, D., Davidov, E., and Schmidt, P. (2026). Level and change of attitudes toward refugees in germany. *International Journal Of Public Opinion Research*, 38(2):edag008.
- Quillian, L. (1995). Prejudice as a response to perceived group threat: Population composition and anti-immigrant and racial prejudice in europe. *American sociological review*, pages 586–611.
- Sauer, C., Auspurg, K., Hinz, T., and Liebig, S. (2011). The application of factorial surveys in general population samples: The effects of respondent age and education on response times and response consistency. In *Survey Research Methods*, volume 5, pages 89–102.
- Scheve, K. F. and Slaughter, M. J. (2001). Labor market competition and individual preferences over immigration policy. *Review of Economics and Statistics*, 83(1):133–145.
- Schiappa, E., Gregg, P. B., and Hewes, D. E. (2005). The parasocial contact hypothesis. *Communication monographs*, 72(1):92–115.
- Schmidt, K., Harth, N. S., and von Harbou, F. (2024). Hierarchien der solidarität. interdisziplinäre betrachtungen zur ungleichbehandlung geflüchteter aus syrien und der ukraine. *German Journal of Forced Migration & Refugee Studies/Zeitschrift für Flüchtlingsforschung (Z’Flucht)*, 8(1):5.
- Schwerter, F. and Zimmermann, F. (2020). Determinants of trust: The role of personal experiences. *Games and Economic Behavior*, 122:413–425.

- Shirmohammadi, M., Beigi, M., and Stewart, J. (2019). Understanding skilled migrants' employment in the host country: A multidisciplinary review and a conceptual model. *The International Journal of Human Resource Management*, 30(1):96–121.
- Sides, J. and Citrin, J. (2007). European opinion about immigration: The role of identities, interests and information. *British journal of political science*, 37(3):477–504.
- Stefaniak, A., Echterhoff, G., Tausch, N., Herzog, M., Groß, J., Thölmann, C., and Becker, J. (2025). Double standards in the treatment of refugees: Why are people more likely to help ukrainian compared to syrian refugees?/dobles estándares en el trato a los refugiados:¿ por qué se tiende más a ayudar a los refugiados ucranianos que a los sirios? *International Journal of Social Psychology*, 40(2-3):219–265.
- Wolff, R. V., Struck, O., and Osiander, C. (2025). Skilled labour immigration: a vignette analysis on the willingness to accept migrants from outside the eu. *Journal for Labour Market Research*, 59(1):14.

## A Appendix Survey Instrument

what to add here: vignette instructions, exact vignette template, all six outcome questions with every response option, contact/experience questions, media questions, general-belief questions, demographics, and attention checks.

Should I also make a survey in Qualtrics and include a link to it here? Or add screenshots of the survey questions which I create in Qualtrics?

This should probably be the largest appendix and contain the material that allows someone to understand exactly what respondents would see.

### A.1 Vignette experiment and profile construction

Include the vignette introduction/instructions, the exact vignette template, and the five randomized attributes and their levels. You can then show perhaps 2–4 example profiles, rather than screenshots of all 32 possible profiles.

This is also the natural place for the Qualtrics screenshots you wanted to include.

### A.2 Outcome questions and response scales

Put the exact wording of all five outcome questions here:

$Y^{\text{Emp}}$ ,  $Y^{\text{Inc}}$ ,  $Y^{\text{Hire}}$ ,  $Y^{\text{Work}}$ , and  $Y^{\text{Soc}}$ .

This directly fulfils the Appendix [X] promise currently made in Section 3.3.

For each question, show the exact response scale/categories. In particular, list all seven income categories rather than merely saying “seven categories.”

### A.3 Personal contact and experience questions

Include exact wording and response scales for workplace contact, non-work contact, overall experience quality, whether a memorable experience comes to mind, and memorable-experience valence.

This fulfils your current Appendix [X] reference in Section 3.4.1.

### A.4 Media exposure and information questions

Even though I don’t see a separate Appendix [X] reference attached to this subsection in the current PDF, I would absolutely include it. Otherwise the appendix would give exact wording for contact but strangely omit the parallel media measures.

Include  $M_{ig}$ ,  $T_{ig}$ ,  $I_{ig}$ , and  $V_{ig}^I$ , including the relevant skip logic.

### A.5 General-belief questions

Include exact wording/scales for all five:

$$TimeLM_i, \quad TimeSoc_i, \quad Comm_{ig}, \quad FacilEmp_{ig}, \quad RedInt_{ig}.$$

Again, this makes the questionnaire reproducible even though your current text does not explicitly send every one of these questions to Appendix [X].

## A.6 Respondent characteristics

Include the exact wording, response categories, and coding of age, gender, education, employment status, household income, region, migration background, religion, and political orientation. This directly fulfils your Section 3.4.4 promise.

# B Appendix Data Processing

## B.1 Data-quality procedures

Briefly operationalize what Section 3.5 already explains: attention check, completion time, duplicate/automated-response indicators where available, and the rule that indicators are considered together rather than automatically excluding someone based on one flag.

Because your thesis already explains this in prose, do not repeat several paragraphs. A concise table will be enough.

## B.2 Missing, not-applicable, and structurally unavailable responses

This is where I would put the table we have been discussing.

A preliminary version could be:

## B.3 Construction of derived variables

This should contain the operational rules for variables that are not simply raw survey answers.

Most importantly, spell out  $C_{ig}$ :

$$C_{ig} = \frac{C_{ig}^{\text{Work}} + C_{ig}^{\text{NonWork}}}{2}$$

when both measures are available, whereas

$$C_{ig} = C_{ig}^{\text{NonWork}}$$

when workplace contact is genuinely not applicable. That is already the rule in your main text.

Also include the construction of

Table 2: Treatment of Missing, Not-Applicable, and Conditional Responses

| Situation                                                                                 | Classification              | Coding and treatment                                                                                                                                             | Consequence for analysis                                                                                                                                   |
|-------------------------------------------------------------------------------------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| No memorable personal experience comes to mind ( $E_{ig} = 0$ )                           | Structurally unavailable    | $V_{ig}^E$ is missing because the valence question is not displayed.                                                                                             | The respondent remains in other analyses. The observation is not used in the H6a specification requiring $V_{ig}^E$ .                                      |
| No memorable media information comes to mind ( $I_{ig} = 0$ )                             | Structurally unavailable    | $V_{ig}^I$ is missing because the valence question is not displayed.                                                                                             | The respondent remains in other analyses. The observation is not used in the H6b specification requiring $V_{ig}^I$ .                                      |
| Respondent has not worked during the relevant period                                      | Not applicable              | $C_{ig}^{\text{Work}}$ is coded as missing rather than as no workplace contact. The overall contact measure $C_{ig}$ is based on $C_{ig}^{\text{NonWork}}$ only. | The respondent remains in the analysis. The unavailable workplace-contact measure does not result in respondent-level exclusion.                           |
| Respondent has worked but reports “never” having workplace contact with migrant group $g$ | Valid substantive response  | $C_{ig}^{\text{Work}} = 1$ , corresponding to “never” on the contact-frequency scale.                                                                            | The response is included normally in the construction of $C_{ig}$ and is not treated as missing.                                                           |
| Respondent selects “Prefer not to say” for a displayed question                           | Explicit nonresponse option | The response is recorded as “Prefer not to say” and coded as missing for the corresponding variable in the analysis.                                             | The respondent remains in analyses that do not require the respective variable. The observation is unavailable for specifications requiring that variable. |

*Notes:* All survey questions displayed to respondents require a response before they can proceed. Where appropriate, particularly for sensitive respondent characteristics, respondents can select “Prefer not to say.” Missing values therefore arise from explicit nonresponse options, questions that are not applicable, or questions that are not displayed because of conditional survey logic, rather than from respondents skipping displayed questions.

$$Positive_{ig} = \max(Q_{ig} - 4, 0)$$

and

$$Negative_{ig} = \max(4 - Q_{ig}, 0),$$

and any other transformations/codings that are not immediately obvious from the raw questionnaire.

## B.4 Analysis-sample availability

This can be very short, perhaps even another small table rather than prose. It would connect the missing-data rules to your empirical strategy:

Table 3: Data Requirements for the Main Analyses

| Analysis                | Additional data required                        |
|-------------------------|-------------------------------------------------|
| H1–H3                   | Valid vignette outcome                          |
| H4a–H4b                 | Relevant contact and experience measures        |
| H5                      | Relevant media exposure and media-tone measures |
| H6a                     | $E_{ig} = 1$ and observed $V_{ig}^E$            |
| H6b                     | $I_{ig} = 1$ and observed $V_{ig}^I$            |
| Adjusted specifications | Required respondent-level control variables     |

*Notes:* The number of observations used in each specification may differ depending on the availability of the variables required for the respective analysis. Missing information for a variable not required by a particular specification does not by itself result in respondent-level exclusion.

## Statement of Authorship

I hereby confirm that the work presented has been performed and interpreted solely by myself except for where I explicitly identified the contrary. I assure that this work has not been presented in any other form for the fulfillment of any other degree or qualification. Ideas taken from other works in letter and in spirit are identified in every single case.

Date:

Signature: